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1. Genetic Algorithm

Given:

Objective function:

$$f(x) = x^2$$

Initial population:

$$\{2, 4, 6, 8\}$$

Step 1: Calculate Fitness Values

Individual (x)	**Fitness** $f(x) = x^2$
2	$2^2 = 4$
4	$4^2 = 16$
6	$6^2 = 36$
8	$8^2 = 64$

Step 2: Selection

The individuals with the highest fitness are selected.

- **8 → Fitness = 64**
- **6 → Fitness = 36**

These are chosen as the parent chromosomes.

Step 3: Crossover

Convert the selected individuals into 4-bit binary.

- **8 = 1000**
- **6 = 0110**

Exchange the least significant bit (LSB).

Parent 1 : 1000

Parent 2 : 0110

Both parents have LSB = **0**, so after crossover:

Offspring 1 : 1000 = 8

Offspring 2 : 0110 = 6

Therefore, the offspring remain unchanged.

Role of Mutation

Mutation randomly changes one or more bits of a chromosome with a small probability.

Importance of Mutation:

- Increases population diversity.
- Prevents premature convergence to local optimum.
- Introduces new genetic information.
- Helps explore new solutions.
- Improves the probability of finding the global optimum.

Final Answer

Fitness Values:

Individual	**Fitness**
2	4
4	16
6	36
8	64

Selected Parents: 8 and 6

Offspring after LSB Crossover: 8 and 6

Mutation: Introduces random changes to maintain diversity and improve the search for the optimal solution.

2. Perceptron Model

Given

- Advertisement Exposure (x_1) = 2

- Income Level (x_2) = 1
- Weight $w_1 = 0.5$
- Weight $w_2 = 0.3$
- Bias $b = -0.4$

Step 1: Calculate Net Input

The perceptron equation is:

$$\text{Net Input} = w_1x_1 + w_2x_2 + b$$

Substitute the given values:

$$\begin{aligned} &= (0.5 \times 2) + (0.3 \times 1) + (-0.4) \\ &= 1.0 + 0.3 - 0.4 \\ &= 0.9 \end{aligned}$$

Step 2: Apply Step Activation Function

The step activation function is:

$$\text{Output} = \begin{cases} 1, & \text{if Net Input} \geq 0 \\ 0, & \text{if Net Input} < 0 \end{cases}$$

Since,

$$0.9 \geq 0$$

Output = **1**

Decision

Output **1** indicates that the customer **will purchase the product**.

Final Answer

- **Net Input = 0.9**
- **Activation Output = 1**
- **Prediction: Customer purchases the product.**